



Between the lines: Data Analysis of our friend group chat

Bas hooge Venterink | Student nr: 1905776

Data Analysis & Visualisation | 05-04-2026

 <https://github.com/bashoogeventerink-max/DAV>

Intro

For the course Data Analysis & Visualisation class, the assignment is to analyze data from a Whatsapp chat. I chose to analyze one of my most used chats: the group chat of my friends which I met in my hometown Losser during high school. The group chat contains nine friends in total, sending roughly ten thousand messages spanning over a time period of 3+ years.

With the aging of our friend group from 25 to 30 during these years, the group members have gone through some significant developments in their lives. Some examples of these are moving away from the hometown, developing new relationships, and moving in with their partner. My personal goal for this assignment was to investigate what the effects are of some of these developments on the dynamics in the group chat of the friend group. Using the following types of visualizations, you can see some of these effects in this report:

- Comparing Categories
- Time series
- Relationship
- Distribution
- Modeling

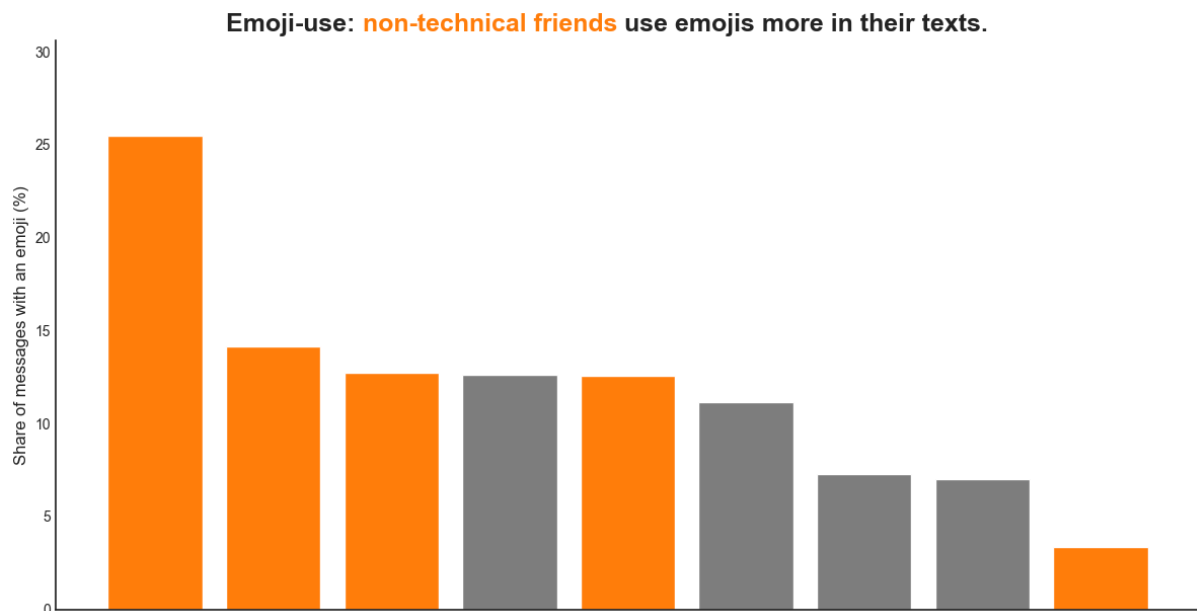
After consent of my friends to use the messages in of the group chat, the data anonymized using the following function:

- ``_anonymize_authors``

Before anonymizing though, several features have been created which can be attributed to each unique friend, for instance if he lives with their partner and if the friend is living in the hometown or not.

After anonymizing, several other features are created such as the response time between messages, usage of emoji's and detecting who initiates meeting up. In the following chapters you can find the visualizations along with context, story, and reflection from my side.

Comparing categories



Context

Over the years, emojis have become an increasing part of our way of communication to express ourselves. This graph shows the percentage of messages containing an emoji (y-axis) per unique user (x-axis), split by whether the user has a technical study background (orange) or not (grey).

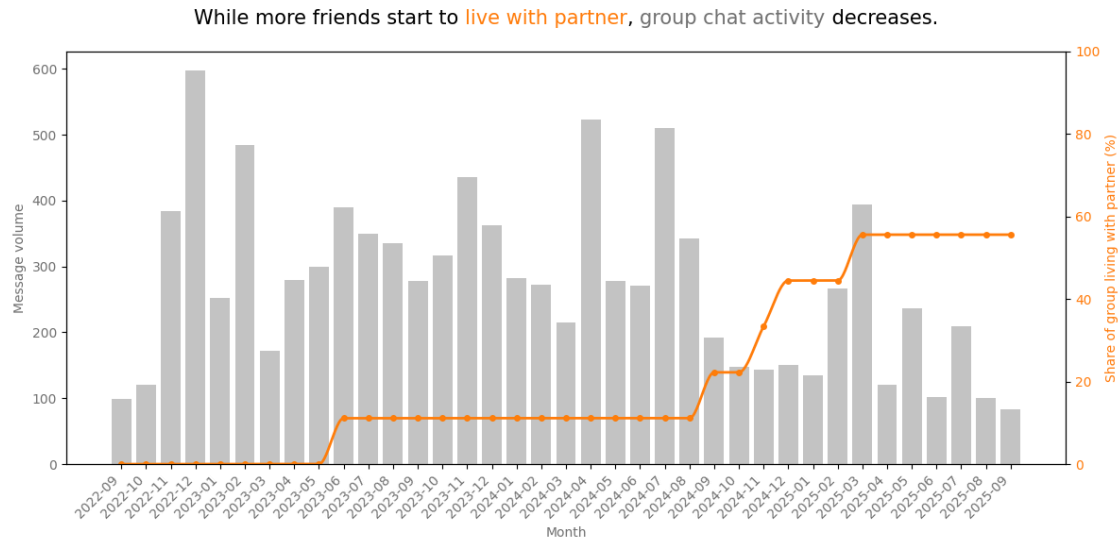
Story

Using a bar chart, we see that non-technical friends are using emojis relatively more often compared to their counterparts. This indicates that non-technical friends express themselves using an emoji in their messages more often.

Reflection

The first element of the graph that draws attention is that the orange bars are generally larger than the grey bars. Another element that you notice is that the bars are sorted descending by percentages, applying the proximity and continuity principles. The principle of similarity is applied by the same color in the title as for the bars for non-technical users. Minimal usage of labels in the graph and on the axis maintain clarity for the reader. Though there seems to be a distinction in emoji-usage by background, the reasons behind using more/less emojis are not proven by this analysis.

Time series



Context

Does living with a partner have an effect on group chat activity? As stated in the intro, the relationships of each of the friends have changed over the years as we are aging from 25 to 30. Over these years more friends have started living with their partner. For the time series graph, I wanted to discover if the trend of friends living with a partner has changed the group activity.

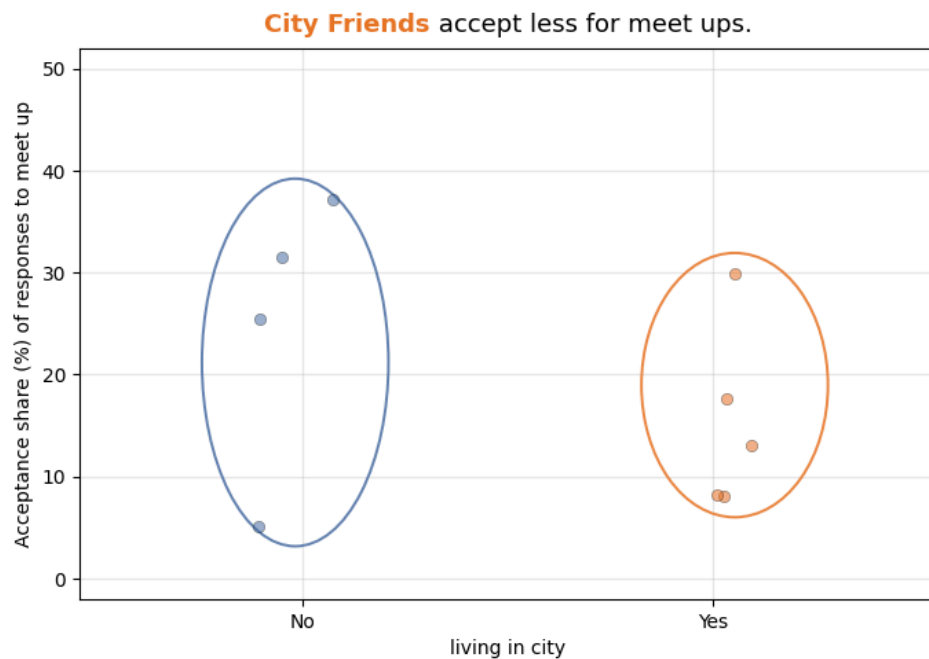
Story

We see that there seems to be a diverging pattern: while the percentage of friends living with their partner increases from the end of 2024 onwards, the overall message volume follows a negative trend in the same period.

Reflection

The first thing you notice is two diverging groups of data over time, indicating contradicting trends. The distinction between these two groups is made by the usage of two unique colors and graph types, applying the similarity principle. By using two x-axis, we solve scaling issues in the datapoints. By using a line graph with connecting points, the closure and connection principles are applied. While the trends are contradictory to each other over time, other factors can apply to the negative trends in message volume as well. Therefore, we cannot fully support the hypothesis that living with a partner decreases the group chat activity.

Relationship



Context

How often do friends accept invites for meet ups? This graph displays the percentages of accepting the invite per user compared to all responses to meet up invites, grouped by whether the city lives in the city or not.

Story

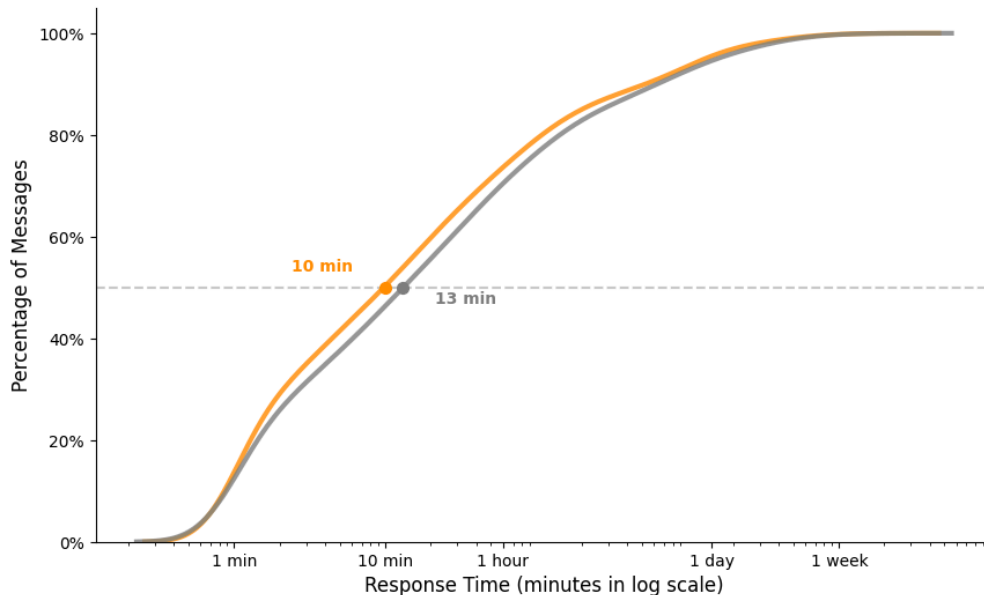
Using a scatterplot, we can see that the percentages of friends living in the city are grouped in a lower percentage group compared to a friend not living in the city, indicating that

Reflection

The first element that pops out is that the datapoints are grouped in two, also indicated by color, applying the proximity and similarity principles. By grouping the dots using a surrounding circle, the enclosure principle is applied. By using a scatterplot, the continuity principle is not applied. Minimal usage of labels and text in the graph maintain clarity for the reader. While we see that most friends living in the city accept relatively less when responding, the relationship is not clearly evident.

Distribution

Friends living in city engage quicker within group chat.



Context

Does city living dictate chat speed? This graph displays the percentage of messages cumulatively distributed over the minutes (log scale). With 55% of the group relocating to a city for work, we are testing if friends in the city, influenced by a faster-paced lifestyle, respond more quickly than their non-city counterparts.

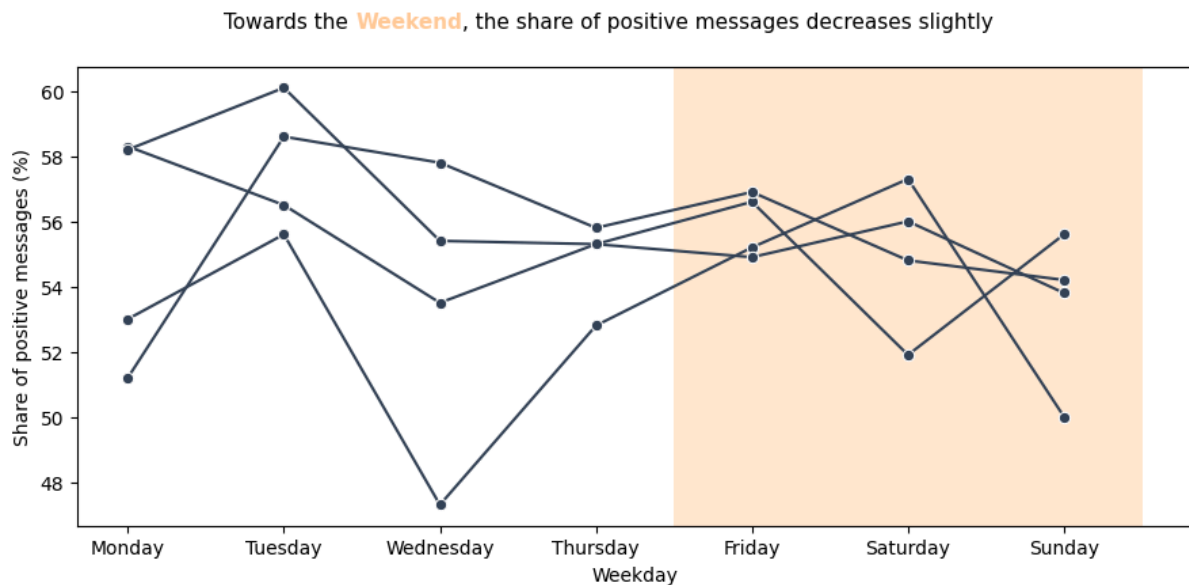
Story

Using a cumulative distribution function, we see that the city group is indeed faster in responding, as the orange line increases faster to the 50th percentile of messages.

Reflection

A highlight of the graph is that the orange line “pops out”, leading to the reader focussing on this line the first time looking at the graph (Preattentive processing). Also, the orange line is faster compared to the grey line, and both lines are increasing to a maximum. These two elements support the principles of similarity and continuity. While the pattern is clear, the reasons behind faster response time are not proven.

Modeling



Context

What sentiment exists within the group chat? This graph displays the share of messages with a positive sentiment. With 55% of the group relocating to a city for work, we are testing if friends in the city, influenced by a faster-paced lifestyle, respond more quickly than their non-city counterparts.

Story

Against my expectations, we see a slightly decreasing trend going towards the weekend. I would expect that the group chat's sentiment would be more positive, for instance as we are happy to meet on the weekend, but there is a slightly decreasing sentiment instead.

Reflection

A highlight of the graph is the decreasing trend in all the lines, except for 2022. By highlighting the weekend in the graph with an orange shade, the enclosure principle is applied. Another element that the reader notices is the color matching between the word "weekend" in the title and the color shade, which support the similarity principle. The connection principle is applied by connecting the datapoints of each day with a line. For clarity, the lines are displayed using a smooth pattern.

Conclusion

These visualizations provide some insights into how the development and aging of the group from 25 to 30 influences group communication. By using a categorical graph, we see that the background of my friends define how much they use emojis in their messaging. Another finding is that the group chat activity has decreased over time, while the percentage of friends living with their partner has increased simultaneously. As a part of the group has moved to the city, we see that they accept less invites for meet-ups compared to people staying in the hometown. Having said that, the friends living in the city do respond faster in the group chat, showing quicker engagement. Lastly, we see a decreasing sentiment in the weekend's messages compared to the weekdays.